

Enterprise Data Engineering, Analytics & AI

Data Modelling Concepts

Fact Tables · Dimension Tables · Star Schema · Snowflake Schema · Modelling Decisions

By the End of This Session...

You will understand how enterprise data is organised for analysis — the architecture behind every BI dashboard you'll ever build



Define Fact & Dimension tables

Understand the roles of each and what kinds of data belong where



Design a Star Schema

The most widely used analytical data model in enterprise BI



Understand Snowflake Schema

When and why to extend Star schema into Snowflake



Apply models to real-life examples

Map retail, banking and healthcare data into fact-dimension models



Make data modelling decisions

Know what questions to ask before designing any model

Every dashboard you've ever seen...

...is powered by a data model built by someone like you. Today we learn how that model is designed.



E-Commerce

"How did each product category perform this quarter, by region?"

Needs: sales data + product info + region info + dates



Banking

"What is the total loan amount disbursed per branch per month?"

Needs: loan data + branch info + time info



Healthcare

"How many patients per doctor per department were seen last week?"

Needs: appointment data + doctor info + department info + dates

What Is Data Modelling?

Data Modelling is the process of defining HOW data is **organised, stored and related** so that it can be queried efficiently for business decisions.

Without a Model — Raw Data

TxnID	Product	Category	Region	SalesRep	Date	Amount	Discount	Tax	Cost
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1001	iPhone 15	Electronics	West	Rahul	2024-01-05	89000	5%	18%	62000
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1002	Shirt	Apparel	North	Priya	2024-01-05	1200	0%	12%	800
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1003	iPhone 15	Electronics	West	Rahul	2024-01-06	89000	5%	18%	62000
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Problems: Repeated data · Slow queries · Hard to update · No clear structure

With a Model — Structured

FactSales

TxnID

DateKey

ProductKey

RegionKey

Amount

Tax

Discount

DimDate

DateKey

Date

Month

Year

DimProduct

ProductKey

ProductName

Category

Cost

DimRegion

RegionKey

Region

Zone

The Fact Table

The centre of every analytical data model — holds the WHAT happened and HOW MUCH



Records business events

Every row = one business event: a sale, a login, a transaction, an appointment, a shipment



Contains measurable metrics

Sales amount, quantity, discount %, tax, profit, duration — values you can aggregate (SUM, AVG, COUNT)



Holds Foreign Keys (FKs)

Links to dimension tables via keys: DateKey, ProductKey, CustomerKey — NEVER stores descriptive text



Grows fast and stays wide

Fact tables have many rows (millions/billions) and a predictable set of columns



'Which column in FactSales would you use to calculate total revenue?'

FactSales — Example

SalesID	INT PK
DateKey	INT FK
ProductKey	INT FK
CustomerKey	INT FK
RegionKey	INT FK
SalesRepKey	INT FK
Quantity	INT
UnitPrice	DEC
Discount	DEC
TaxAmount	DEC
TotalAmount	DEC



Metric columns



Foreign key columns (FK)

Dimension Tables

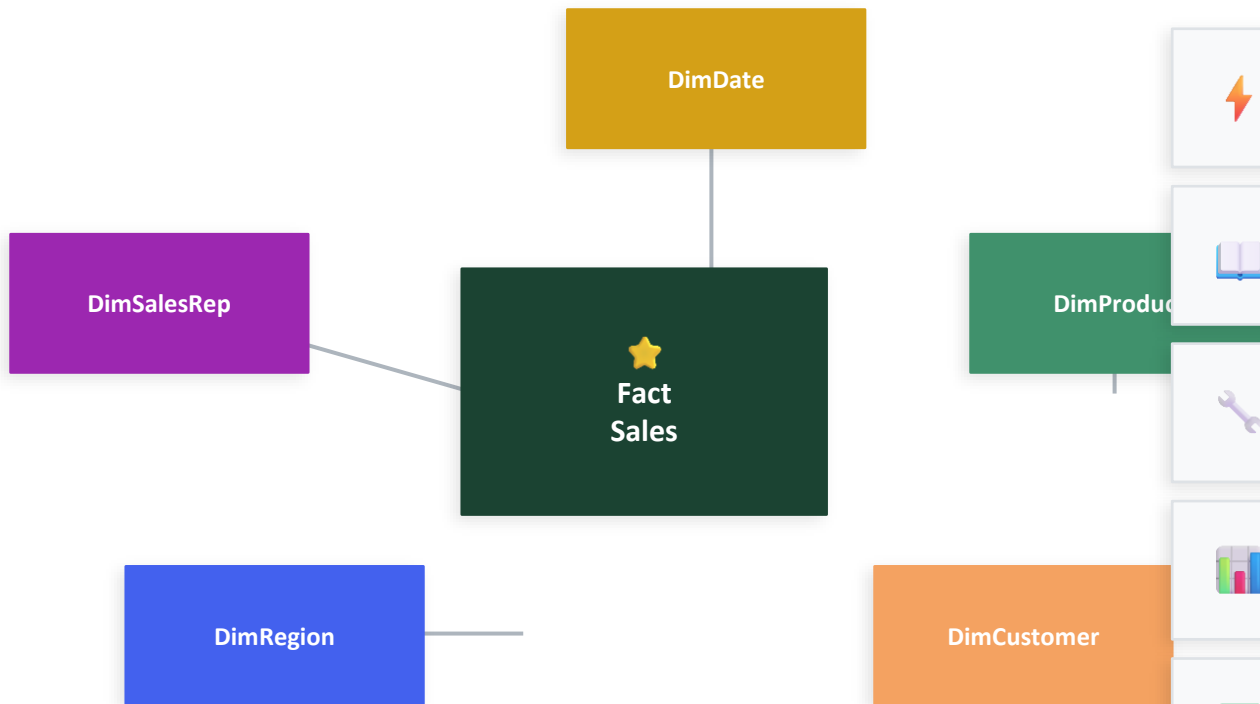
The context providers — they answer WHO, WHAT, WHERE, WHEN around the facts

				
WHEN?	WHAT?	WHO?	WHERE?	BY WHOM?
DimDate	DimProduct	DimCustomer	DimRegion	DimSalesRep
DateKey PK	ProductKey PK	CustomerKey PK	RegionKey PK	RepKey PK
Date	ProductName	FullName	Region	RepName
Day	Category	Email	State	Team
Month	SubCategory	City	Country	Manager
Quarter	Brand	Segment	Zone	Joined
Year		Age		

🔑 Rule: Dimension tables are SMALL & WIDE (few rows, many descriptive columns). Fact tables are TALL & NARROW (many rows, few columns — mostly keys and metrics).

Star Schema

One Fact table at the centre · Dimension tables radiating outward · Looks like a star



Why Star Schema?



Fast query performance — minimal joins needed



Easy to understand — even for non-technical users



Simple to maintain — add a new dimension without redesigning



Native to BI tools — Power BI, Tableau optimise for it

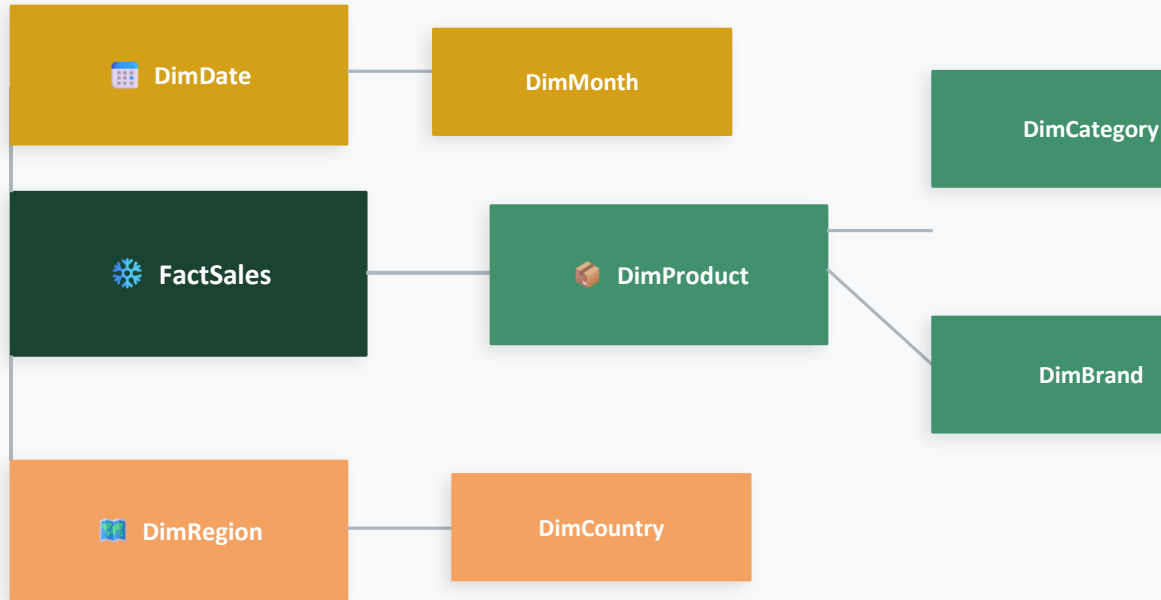


Industry standard — used in 80%+ of enterprise DWs

draw a line from FactSales to a dimension of their choice and explain why.

Snowflake Schema

An extension of Star Schema — dimension tables are further normalised into sub-dimensions



Star vs Snowflake — Key Differences

	★ Star	❄ Snowflake
Dim tables	Denormalised	Normalised
Joins	Fewer (faster)	More (slower)
Storage	Uses more	Uses less
Complexity	Simple	More complex
Best for	BI / Dashboards	Large DW, storage savings

💡 In which scenario would you choose Snowflake over Star?



QUIZ TIME

3 Questions · 4 Minutes · Use the Chat to submit your answers!

Q1 → What is the key difference between a Fact table and a Dimension table?

Think: numbers vs. context · tall vs. wide · metrics vs. descriptors

Q2 → In a Star Schema for a hospital, name three Dimension tables and their primary keys.

Think: WHEN, WHO, WHERE, WHAT around a medical appointment

Q3 → A company's BI tool is running slowly. A consultant says 'your Snowflake Schema has too many joins.' What would they recommend instead?

Think: which schema has fewer joins and why

Real Life: E-Commerce Star Schema

Business question: Which product category generated the most revenue by region this quarter?

★ FactOrderItem	
OrderItemID	INT PK
OrderID	INT
DateKey	INT FK →
ProductKey	INT FK →
CustomerKey	INT FK →
RegionKey	INT FK →
Quantity	INT
UnitPrice	DEC(10,2)
Discount	DEC(4,2)
Revenue	DEC(12,2)

📅 DimDate
DateKey PK
OrderDate
Day
Month
Quarter
Year

📖 DimRegion
RegionKey PK
Region
State
Country

📦 DimProduct
ProductKey PK
ProductName
Category
Brand
Cost

👤 DimCustomer
CustomerKey PK
Name
Email
City
Segment

💡 How this model answers the question:

```
SELECT p.Category, r.Region,
       SUM(f.Revenue) AS TotalRevenue
FROM FactOrderItem f
     JOIN DimProduct p ON f.ProductKey=p.ProductKey
     JOIN DimDate d ON f.DateKey=d.DateKey
     JOIN DimRegion r ON f.RegionKey=r.RegionKey
WHERE d.Quarter = 'Q1'
GROUP BY p.Category, r.Region
```

Data Modelling Decisions

The questions every data modeller must answer BEFORE designing a schema

01

What is the business question?

Every model serves a purpose. Design backwards from the question — not from the data.

→ 'Which branch had the highest loan defaults?' → you need *BranchKey*, not *BranchAddress* in the Fact

04

Which attributes go in Fact vs Dimension?

Metrics and foreign keys → Fact. Descriptive, categorical, rarely changing data → Dimension.

→ *ProductPrice at time of sale* → Fact (it measures).
ProductCategory → Dimension (it describes).

02

What is the grain of the Fact table?

Grain = the lowest level of detail one row represents. Getting this wrong corrupts every metric.

→ *Order grain: 1 row per order. Line-item grain: 1 row per product per order. They answer different questions.*

05

How will the model handle change over time?

Slowly Changing Dimensions (SCD) — what happens when a customer moves city or changes segment?

→ *SCD Type 1: overwrite. Type 2: add new row with dates. Type 3: add 'Previous' column. (Advanced — Week 2 deep dive).*

03

Star or Snowflake?

Star = simpler, faster queries. Snowflake = normalised, saves storage. Choose based on query needs.

→ *BI Dashboard* → Star. *Large enterprise DW with thousands of product attributes* → Snowflake.

06

Who will query this model?

Analysts using SQL need star schema. Data scientists using Python need different structure. Know your consumer.

→ *Finance team using Power BI* → optimise for DAX. *ML team using Databricks* → optimise for Parquet on Delta Lake.

Session Summary

1

Data modelling organises data into structures that can be queried efficiently for business decisions

2

Fact tables hold WHAT happened — measurable events, metrics and foreign keys. They grow tall and fast.

3

Dimension tables hold CONTEXT — WHO, WHAT, WHERE, WHEN. They are small, wide and descriptive.

4

Star Schema: one Fact at centre, dimensions radiate out. Simple, fast, BI-friendly. Industry default.

5

Snowflake Schema normalises dimensions into sub-tables — more joins, less storage duplication.

6

Great modelling starts with questions: What is the business question? What is the grain? Who queries?

Questions?

*"A well-designed data model is invisible.
A poorly-designed one is unforgettable."*



Before W2 S2 — Think about this:

Pick any app you use daily — Swiggy, your bank, an HR portal.
Imagine you had to build a dashboard for it.
What would your FactTable be? Name three Dimensions.
Share in the group chat at the start of the next session.